Expert Systems With Applications 314 (2026) 131686

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

SCRNet: A spatial-channel reconstruction network for multi-task pineapple detection and a novel pineapple dataset

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ARTICLE INFO

Keywords:
Smart agriculture
Deep learning
Multi-task object detection
Pineapple dataset

ABSTRACT

Accurate localization and maturity assessment of pineapples are essential for achieving fully automated harvesting. However, existing approaches address these two tasks separately, which constrains the overall efficiency of harvesting systems. Due to the high similarity of visual features and frequent occlusions among pineapples, effectively integrating these tasks and performing real-time detection remain challenging. Furthermore, the lack of publicly available datasets hinders algorithm optimization and impedes further research development. In this study, we propose a multi-task visual detection framework for jointly conducting the localization and maturity assessment of pineapples. Consequently, we present a high-quality dataset that includes various maturity degrees, lighting conditions, occlusion situations, and viewing angles, capturing the complexity of real-field environments. This dataset provides a reliable benchmark for developing and evaluating multi-task detection and maturity estimation algorithms. At the same time, we design the Spatial-Channel Reconstruction Network (SCRNet) for joint detection and maturity analysis. First, by refining the backbone architecture and neck structure of a lightweight single-stage detector, SCRNet improves feature extraction and fusion while maintaining low parameter complexity and inference cost, making it well-suited for edge deployment. Second, we design the C3-SCR module, which incorporates an augmented ScConv unit to increase sensitivity to maturity-related features. This design enables the model to achieve accurate pineapple localization and maturity assessment under complex field conditions. Experiment results demonstrate that SCRNet substantially outperforms existing methods in both the localization and maturity assessment. The introduced benchmark dataset not only validates the robustness and generalization of our approach in complex agricultural settings but also enables future advancements in the field. The dataset and code will be made available at: https://github.com/Tuilphenry/SCRNet.

1. Introduction

Pineapple, as one of the major tropical fruits, has shown a significant increase in both cultivation area and market demand worldwide (Mohd Ali et al., 2020). For a long time, pineapple harvesting has primarily relied on manual labor, which not only involves high physical demands and harsh working conditions but also suffers from low efficiency and high fruit loss rates. Due to the growing labor shortage and rising labor costs, the traditional manual harvesting model is increasingly unable to meet the modern agricultural industry's high demands for efficiency, cost-effectiveness, and quality. With the rapid advancement of computer vision technology, automated harvesting techniques based on image recognition and intelligent decision-making are becoming increasingly effective solutions for enhancing pineapple production efficiency, reducing reliance on manual labor, and driving the industry's intelligent transformation (He et al., 2024).

Currently, the application of agricultural automation technology in fruit harvesting has become a significant research focus in modern agrarian engineering. Computer vision technology, serving as the critical link for perception and decision-making in automated harvesting systems, is progressively becoming the core driving force behind intelligent fruit harvesting (Chen et al., 2024a; Zhang et al., 2024a). Advances in deep learning-based algorithm optimization have continuously enriched the technical approaches in computer vision, enabling better adaptation to increasingly complex application scenarios (Kamilaris & Prenafeta-Boldú, 2018; Lin et al., 2025b). In agricultural settings, where real-time decision-making and computational efficiency are critical, lightweight and fast inference models have become increasingly favored. The YOLO

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https://doi.org/10.1016/j.eswa.2026.131686
Received 18 November 2025; Received in revised form 20 January 2026; Accepted 11 February 2026
Available online 14 February 2026
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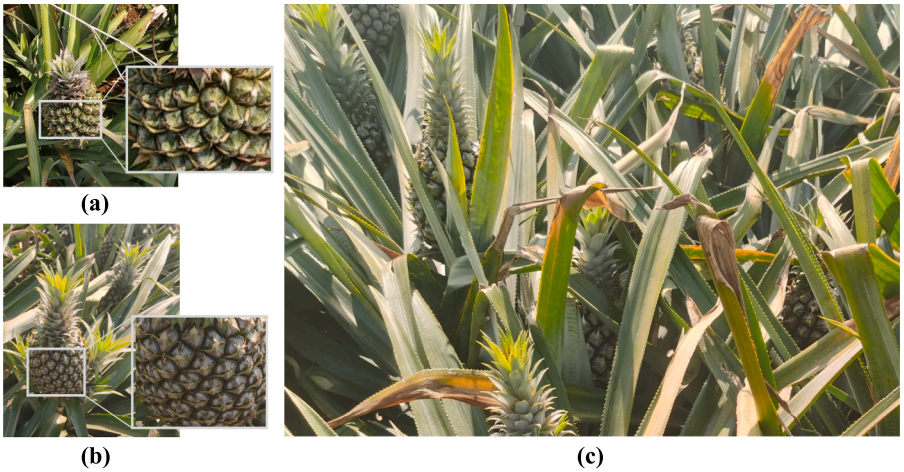

Fig. 1. Challenges in the simultaneous localization and maturity assessment of pineapples. Field appearance comparison of unripe pineapples (a) and ripe pineapples (b). The magnified windows highlight typical discriminative regions. (c) Object occlusion and interference in natural farmland.

(You Only Look Once) series is a widely adopted single-stage object detection framework that performs detection in a single forward pass, offering high speed and good accuracy. Its efficiency and real-time capability make it especially well-suited for fruit detection in complex and dynamic agricultural environments (Deng et al., 2025; Redmon & Farhadi, 2018; Zhu et al., 2024). Consequently, YOLO-based algorithms have been extensively applied in pineapple research. Chang et al. (2022) proposed the TX2-PineNet model based on lightweight convolutional neural networks, capable of real-time maturity classification of bagged pineapples with a 99.27% accuracy under shooting distances of 300-800 mm, arbitrary sunlight conditions, and paper bag coverage. Hong et al. (2025) achieved precise detection of green-ripe pineapples and assessed their maturity by analyzing the withering rate of floral bracts at the fruit base. Lin et al. (2025a) proposed the DPD-YOLO algorithm, which improves the YOLOv8 network by incorporating an attention mechanism to enhance pineapple detection performance under complex environmental conditions. Chen et al. (2025) developed the LTHRNet model based on an improved LiteHRNet, which can accurately detect keypoints and postures of pineapples under varying illumination and occlusion. Shan et al. (2025) introduced MAISNet, which not only detects pineapples but also accurately predicts their rotation angle, thereby effectively improving the perception of fruit orientation for harvesting robots.

Although notable progress has been made in pineapple detection using deep learning algorithms, practical challenges remain for automated harvesting. Most existing studies treat fruit localization and maturity assessment as separate tasks, causing robots to adopt a non-selective harvesting strategy that collects all visible fruits regardless of maturity. As a result, a closed-loop process of "positioning-maturity assessment-selective harvesting" cannot be realized. This limitation necessitates additional manual or automated sorting, increasing labor demands and costs that could otherwise be avoided. Consequently, the economic benefits and operational efficiency of automated harvesting are significantly reduced. Moreover, current maturity assessment methods are predominantly applied post-harvest or to individual pineapples. This restriction arises from their reliance on subtle features such as eye color and skin texture, which are highly susceptible to interference in complex field environments and impose greater demands on model robustness. Generic single-stage detectors, particularly the widely adopted YOLO framework, lack the specialized architectural mechanisms required to address these agricultural challenges. This limitation stems primarily from two inherent structural constraints within the standard YOLO architecture. First, the reliance on isotropic convolution operations creates a significant bottleneck. Due to the content-agnostic nature of the convolution kernel, the model processes texture-rich fruit surfaces and complex backgrounds, such as serrated leaves and ground shadows, indiscriminately. This characteristic introduces substantial spatial redundancy, thereby compromising boundary localization in heavily occluded scenarios. Second, although recent YOLO variants have attempted to enhance feature extraction by integrating attention mechanisms (e.g., SE or CBAM), they often fail to capture complex discriminative features due to suboptimal integration strategies (Usman et al., 2025a). These mechanisms typically employ a 'feature re-weighting' strategy, which globally modulates features based on statistical descriptors. Given the high morphological and textural similarity between unripe and ripe pineapples, simple re-weighting tends to simultaneously amplify both subtle maturity cues and redundant environmental noise. Consequently, this coupled amplification fails to disentangle the fine-grained details necessary for precise maturity classification, necessitating a paradigm shift from passive re-weighting to active feature reconstruction. In addition, the persistent lack of standardized datasets for pineapple detection has constrained both research progress and algorithmic optimization in this domain. Overall, the simultaneous achievement of accurate localization and maturity assessment of pineapples presents several practical challenges:

(i) Subtle feature differences. As illustrated in Fig. 1, the primary challenge arises from the extremely subtle differences between ripe and unripe fruits, mainly reflected in subtle, localized morphological attributes such as fruit-eye distribution, color, skin texture, and flesh firmness. These features are not prominent in conventional images. They are easily affected by environmental factors such as shooting angle, illumination, and occlusion, leading to unstable and low-robustness recognition by mainstream CNNs in maturity assessment tasks.

(ii) Object occlusion and interference in complex environments. The complex and variable growing environments of pineapples, where fruits are often heavily obscured by dense foliage and plant structures, create blurred boundaries between fruits and their backgrounds. Additional interference from weeds, soil, and tool shadows further complicates object extraction and boundary determination, imposing greater demands on the perception capabilities of detection models.

(iii) Limitations of YOLO in pineapple detection. Current single-stage detectors are constrained by isotropic convolutions and coarse-grained attention mechanisms: the former cannot differentiate fruit from complex backgrounds, undermining localization robustness under occlusion; the latter, relying solely on feature reweighting, fails to disentangle the highly similar textural patterns between unripe and ripe pineapples, hindering maturity discrimination.

(iv) The lack of relevant datasets. Unlike apples and citruses, which benefit from abundant publicly available datasets (Häni et al., 2020, James et al., 2024), pineapples lag far behind in dataset

construction. On the one hand, pineapples are cultivated in limited regions, lacking broad geographical coverage (Li et al., 2022). On the other hand, precise maturity annotation is highly labor-intensive and requires professional expertise. Consequently, the long-term shortage of standardized and diverse datasets severely restricts algorithm training depth, model optimization, and the broader application of automated harvesting technologies.

To address these challenges, this study proposes a multi-task object detection model, the Spatial-Channel Reconstruction Network (SCR-Net), which simultaneously performs pineapple localization and maturity assessment, and further provides a dedicated pineapple dataset. The core innovation lies in the novel C3-SCR module, which replaces standard bottlenecks to transition from passive feature re-weighting to active feature reconstruction. By unifying the Spatial Reconstruction Unit (SRU) and Channel Reconstruction Unit (CRU) into a cohesive framework, this module collaboratively addresses the dual challenges identified above. Specifically, the SRU actively suppresses background redundancy (e.g., dense foliage) via spatial filtering to resolve localization ambiguity in occluded scenarios, while the CRU recalibrates fine-grained feature dependencies to disentangle subtle maturity cues from environmental noise. This holistic design ensures that both localization and maturity assessment are optimized within a single efficient architecture. Additionally, we construct a new pineapple dataset to advance research in this field. The dataset is collected from real farmland environments under multiple complex scenarios, covering different maturity stages, illumination conditions, occlusions, and growth variations, and is meticulously annotated by hand. This effort partly alleviates the scarcity of publicly available pineapple maturity datasets. Systematic experimental validation establishes the effectiveness of the proposed method and highlights its promising potential for practical deployment in robotic harvesting applications.

In particular, our main contributions are as follows:

(1) We propose a pineapple dataset collected in natural farmland for joint detection and maturity estimation, covering diverse lighting, occlusion, and capture angles.
(2) We design a multi-task framework integrating the self-designed C3-SCR module. This unified module synergizes spatial filtering and channel reconstruction to simultaneously overcome dense occlusion and subtle visual ambiguity, balancing accuracy and efficiency.
(3) Experiment results show that our method simultaneously achieves optimal positioning accuracy and maturity assessment performance under complex field conditions, underscoring its strong overall performance and real-world applicability.

2. Related work

In recent years, deep learning-driven automated agricultural systems have become a prominent research focus. This technology demonstrates significant applicability across multiple domains, including precise fruit localization (Wang et al., 2022), yield estimation (Attri et al., 2023), pest and disease detection (Gülmez, 2025), and maturity assessment (Rizzo et al., 2023). In the context of automated harvesting, accurate fruit localization and maturity assessment represent the core technical components. This work primarily reviews recent advances in deep learning applied to fruit detection, maturity assessment, and pineapple detection research.

2.1. Fruit detection

Early research on fruit detection primarily relied on traditional image processing techniques, such as detection based on color, shape, and texture features (Gongal et al., 2015). These methods manually design features to differentiate fruits from the background, commonly employing color space conversions, thresholding, and edge detection. However, such approaches exhibit significant limitations in complex orchard environments, including illumination variations, color similarity between fruits and background, and occlusion by leaves, resulting in insufficient detection accuracy and robustness. With the advancement of deep learning, convolutional neural network-based methods have gradually become the mainstream for fruit detection (Chuquimarca et al., 2024).

In traditional image-based fruit detection, Sabzi et al. (2018) employed multi-color space and texture features for apple segmentation, achieving satisfactory performance under diverse lighting conditions. Nyarko et al. (2018) proposed a fruit recognition method based on the detection and classification of convex surfaces in RGB-D images, achieving efficient and accurate fruit identification by designing a computationally efficient Convex Template Instance (CTI) descriptor. However, their method exhibited limited adaptability to complex backgrounds. To overcome this drawback, Fu et al. (2019) developed a detection approach that combined HSV-based color thresholding, texture analysis, and morphological processing, thereby improving robustness in natural environments. Despite these improvements, the method still depended on fixed thresholds, which constrained its generalization ability. To address this issue, Wu et al. (2020) combined color and geometric features for fruit segmentation, achieving improved accuracy and robustness in complex environments. Compared with earlier methods, this framework demonstrated greater robustness under complex conditions. Nonetheless, traditional detection techniques relying on color and texture features remain vulnerable to occlusion, illumination changes, and background complexity.

To address detection challenges in complex environments, Liu et al. (2023a) combined binocular cameras and LiDAR to achieve real-time detection and accurate localization of fruits in complex scenarios. Bargoti and Underwood (2017) adopted the Faster R-CNN framework, leveraging a cascaded structure of region proposal networks and classification and regression, significantly improving detection robustness in complex scenes and achieving accurate fruit detection and spatial localization in natural orchard environments. The two-stage detection model achieves high-precision detection by first performing coarse-grained screening, followed by fine-grained classification that leverages local and detailed features. However, its high latency, high computational demands, and high memory requirements make deployment on edge devices challenging. As a single-stage detection framework, the YOLO series effectively addresses these issues. Gao et al. (2024) proposes LACTA, a lightweight and efficient algorithm that achieves high-accuracy cherry tomato detection in complex greenhouse environments through adaptive feature extraction and cross-layer fusion mechanisms. He et al. (2025) proposed an improved YOLOv8 model and combined it with 3D point clouds to achieve precise identification and pose estimation of broccoli in complex field environments.

2.2. Maturity assessment

Fruit maturity assessment plays a critical role in agricultural production and postharvest processing, as it can optimize harvest timing, improve processing efficiency, and enhance product quality. Traditional maturity assessment mainly relies on subjective human judgment, which is often inefficient and inaccurate. In recent years, considerable research has been conducted in this field, achieving significant progress. Early work primarily used hyperspectral or multispectral imaging to detect internal fruit composition or reflectance characteristics. Zhou et al. (2021) utilized drones in conjunction with near-ground hyperspectral imaging to classify strawberry maturity, demonstrating the capability of hyperspectral data to capture internal fruit characteristics. Nevertheless, this approach demands high image resolution, which restricts its practical flexibility. To address this limitation, Aherwadi et al. (2022) applied multispectral imaging to improve banana maturity classification by optimizing band selection and coverage, thereby reducing dependency on high-resolution equipment. More recently, Zahidi et al. (2024) introduced a dual-band feature selection strategy that extracts reflectance information from key spectral bands, simplifying data processing and enhancing computational efficiency.

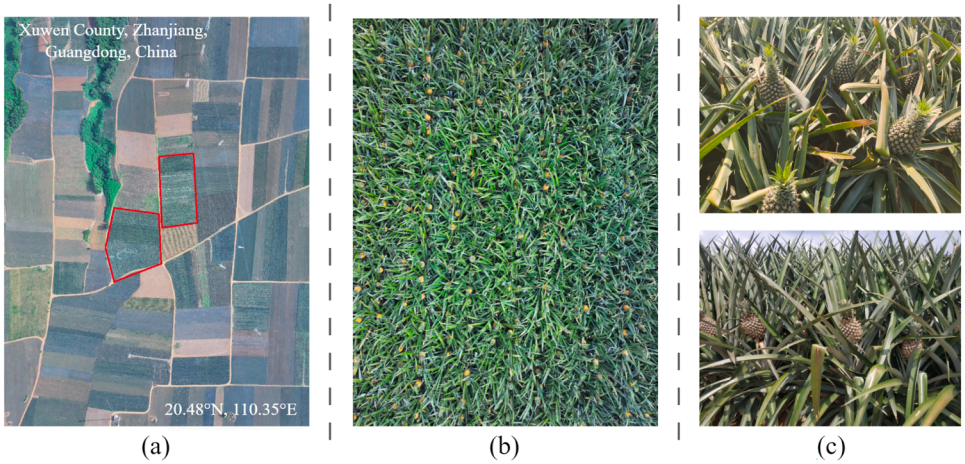

Fig. 2. One of experimental orchards used in this study. (a) Geographical location of the orchard. (b) Orchard captured by UAV at low altitude. (c) Pineapple images from the robotic arm perspective.

Although accurate, such methods require expensive and bulky equipment, making them unsuitable for field deployment. Wang et al. (2024a) developed a YOLOv8-based system for strawberry maturity recognition, combining HSV color channels with ECA attention mechanisms, significantly improving detection accuracy while maintaining real-time performance. To further reduce model complexity for mobile deployment, Yang et al. (2025) proposed MFD-YOLO based on the YOLOv7-tiny framework, introducing multi-scale attention and feature fusion mechanisms to achieve rapid recognition across different strawberry growth stages. Wu et al. (2025) proposed a lightweight RT-DETR model based on the Transformer framework for tomato maturity recognition, incorporating RepBlock and EMA attention mechanisms to balance accuracy and inference speed. Yang et al. (2024) designed the ELD model with Ghost-Shuffle modules and multi-channel attention mechanisms, maintaining high-precision detection under complex occlusion scenarios. Wang et al. (2026) introduced FDFR-Net, a multimodal fusion network that achieves high-accuracy, non-destructive mango ripeness grading by integrating RGB images, hyperspectral data, and impedance features through a two-stage feature-and-decision fusion strategy.

2.3. Pineapple detection

Research on pineapple detection primarily focuses on two key tasks: localization and maturity assessment. To address challenges in complex agricultural environments, early studies integrated advanced modules into detection frameworks. Meng et al. (2023) combined region-based CNNs with shifted window attention mechanisms, effectively enhancing robustness under variable lighting. However, the high computational demand of such spatiotemporal models limits their deployment on edge devices. To mitigate this, Zhang et al. (2024b) introduced the lightweight MobileOne network into YOLOv7 (MSGV-YOLOv7), significantly reducing model complexity. Nevertheless, precise positioning in cluttered backgrounds remains a challenge. To further enhance localization accuracy, Li et al. (2025) incorporated the CBAM attention module into YOLOv7-tiny, leveraging channel weighting to strengthen feature representation. Beyond algorithmic improvements, Liu et al. (2023b) combined binocular stereo vision with an industrial camera array to achieve millimeter-level localization accuracy, advancing robotic harvesting applications.

Regarding maturity assessment, Chang et al. (2022) pioneered the deployment of a YOLO-based system on the NVIDIA TX2 platform, verifying the feasibility of real-time recognition using spectral features. However, standard detection models often struggle to capture subtle, local textural details critical for maturity assessment. To address this limitation, Lai et al. (2023) enhanced the YOLOv7 framework by integrating the SimAM attention mechanism and optimizing bounding box regression with soft-NMS. This channel-weighting strategy significantly improved feature capture, achieving 92.00% accuracy for ripe pineapples. In contrast to these bounding box regression methods, Nasution et al. (2023) employed instance segmentation combined with BlendMask for pixel-level maturity grading. While this segmentation-based approach provides detailed mask information, the substantial computational overhead inherent to such complex networks severely constrains their real-time processing capabilities on resource-constrained edge devices.

2.4. Multi-task learning approaches in agricultural

Early research in the agricultural domain primarily prioritized optimizing Convolutional Neural Networks (CNNs) for single-task efficiency. For instance, Jiang et al. (2019) and Faisal et al. (2020) developed efficient deep learning models for apple leaf disease detection and date palm maturity classification, respectively. While these single-task approaches yielded significant performance gains, independent training regimes result in computational redundancy and fail to exploit potential feature complementarity between localization and attribute discrimination.

To achieve efficient integrated perception within a unified framework, Zhang et al. (2024d) introduced a Multi-Task Learning (MTL) paradigm that simultaneously optimizes freshness detection and variety classification via a shared CNN backbone. Addressing the trade-off between detection speed and multi-task accuracy, Chen et al. (2024b) proposed MTD-YOLOv7. By employing decoupled decoder branches and the SIoU loss function, this model facilitates the simultaneous detection of individual cherry tomatoes and the assessment of bunch maturity. To further enhance robustness against scale variations, Wang and Luo (2024) integrated attention mechanisms into multi-task recognition. Their proposed Multi-Scale Attention Network (MSA-Net) amplifies feature responses to fruits of varying sizes, effectively mitigating multi-scale feature alignment challenges in complex backgrounds. More recently, to improve generalization in unstructured environments, Zhu et al. (2025) investigated the utility of foundation models. Leveraging the zero-shot transfer capabilities of Grounding-DINO, this approach achieved robust apple ripeness and size estimation, demonstrating the potential of large-scale pre-trained models in selective harvesting tasks.

Table 1
Detailed statistics of the constructed pineapple dataset.

Type	Minimum	Maximum	Average
Image width (pixels)	1280	4032	3563.38
Image height (pixels)	690	4032	3023.35
Collection time (hh:mm)	8:30	17:30	10:30
Shooting height (m)	1.2	3.0	1.8
Total number of unripe pineapples	–	13,098	–
Total number of ripe pineapples	–	1825	–
Unripe pineapples per image	0	73	10.41
Ripe pineapples per image	0	15	1.38
Object width (pixels)	16.0	2045.0	224.0
Object height (pixels)	26.0	2400.0	333.8

3. Materials and methods

3.1. Pineapple dataset

3.1.1. Data details

As shown in Fig. 2, the data collection for this study is conducted in Xuwen County, Guangdong Province, PRC. To comprehensively characterize pineapple fruit development under real-world agricultural conditions, we collected unharvested samples across both mature and immature growth stages. Despite focusing on Xuwen County, China's largest pineapple-producing region, we systematically sampled five representative orchards featuring distinct topographic characteristics and planting densities. Sampling was conducted across winter and spring to capture fruits under diverse environmental conditions, ensuring the dataset's representativeness and reliability for agricultural applications. While the data collection is geographically concentrated, Xuwen represents a typical tropical pineapple cultivation environment. Therefore, the intended scope of generalization for the proposed method focuses on agricultural scenarios with similar climatic conditions and planting patterns, serving as a reliable reference for deployment in comparable tropical or subtropical regions.

A high-resolution industrial camera is employed for data acquisition, mounted on either a tripod or a handheld device, and evenly distributed across the planting area to capture images from different orientations, angles, and heights. All images are taken under natural lighting conditions and include diverse scenarios of illumination, shadow, and occlusion. During acquisition, researchers manually adjust the field of view to ensure that the number of pineapples, their distribution density, and typicality in each image meet the diversity requirements. Some scenes feature severe leaf occlusion and overlapping pineapples, thereby enhancing the dataset's ability to simulate the complexity of robotic harvesting scenarios. To improve representativeness and scientific value, all collected images are manually pre-screened to remove low-quality samples, such as blurred or overexposed images. The final dataset comprises a total of 1295 images, encompassing pineapple fields with different planting densities, lighting conditions, and growth stages. This dataset, with its rich diversity and realistic complexity, provides a solid foundation for the development and evaluation of multi-task detection and maturity assessment algorithms.

3.1.2. Image annotation

The dataset annotation process is jointly guided by agricultural experts and computer vision specialists and carried out by experienced annotators. All annotators are familiar with pineapple orchard imagery, have a solid understanding of the pineapple detection task, and follow annotation guidelines strictly developed by domain experts. To guarantee annotation quality and mitigate subjectivity, we implemented a rigorous cross-checking workflow. Each image was initially labeled by one annotator and subsequently reviewed by a second independent annotator. In cases of disagreement, particularly for borderline samples where the transition from concave to flat eyes was ambiguous, the data were escalated to a senior agricultural expert for a final decision. This "Label-Review-Arbitrate" mechanism ensures strict expert consensus, significantly improving the consistency and reliability of the dataset.

For each pineapple fruit, annotation is performed using four coordinate points along with a maturity category $(x_{\min}, y_{\min}, x_{\max}, y_{\max}, class)$ to define its position in the image and to indicate whether it is mature. The bounding boxes cover the actual fruit size; even in cases of partial occlusion by leaves, the occluded regions are included in the annotated range to ensure a complete target representation. The pineapple samples in this dataset are categorized into two groups: ripe and unripe. This binary classification strategy is primarily driven by the decision-making constraints of the robotic harvesting system, which fundamentally requires a clear "Harvest" (actionable) or "Wait" (non-actionable) command. Modeling maturity as a binary task aligns with this operational logic, ensuring high decision efficiency for the robotic arm. While multi-stage maturity grading could offer deeper agronomic insights for yield management, it is considered a direction for future work. The biological threshold for maturity assessment is based on morphological changes in the fruit: unripe pineapples exhibit a distinctly concave or sharply angled contour around the pineapple eyes, indicating the flesh has not yet fully expanded. When the contour around the pineapple's eyes transitions from concave to rounded or nearly flat, the fruit has reached the maturity standard for harvest. This change is a direct morphological manifestation of the internal flesh cells fully swelling and accumulating sugar content. To guarantee annotation quality, each image is initially labeled by one annotator and subsequently reviewed by another, with this double-checking mechanism significantly improving annotation consistency and accuracy. Statistical analysis shows that the dataset contains approximately 15,000 pineapple instances. The bounding box annotation and maturity assessment tasks are performed by about five annotators over a period of 30 days.

3.1.3. Dataset analysis and characteristics

To ensure the generalization capability of the model and strictly prevent data leakage, we implemented a rigorous methodology for dataset construction and analysis. Although the data collection was concentrated in Xuwen County, sampling was conducted across five distinct experimental plots with varying terrain and planting densities to maximize environmental representativeness. The detailed statistics of the constructed dataset are summarized in Table 1.

To avoid the data leakage issue common in random splitting, in which adjacent frames of the same plant might appear in both training and testing sets, we adopted a scene-based stratified splitting strategy, referencing the data construction methodology presented by Khan et al. (2024a). Images captured from continuous collection paths were grouped as single units and assigned exclusively to training, validation, or testing sets in a 7:2:1 ratio. This ensures that the model is evaluated on unseen plants and viewing angles, guaranteeing a fair assessment of robustness.

To substantiate the diversity claims, we performed a quantitative statistical analysis focusing on illumination and scene complexity. As shown in Fig. 3, to ensure robustness against field lighting changes, data collection was conducted across multiple time periods (spanning from 8:30 to 17:30) and varying weather conditions. Following rigorous manual screening to remove low-quality samples, the resulting pixel intensity distribution exhibits a broad Gaussian-like curve with $\mu = 120.0$ and a significant standard deviation of $\sigma = 26.8$. The data effectively covers a dynamic range of $[50, 185]$. This wide span confirms that the dataset captures diverse natural lighting scenarios-from morning shadows to bright noon daylight. Further statistical analysis confirms the dataset's complexity. The object scales cover a broad range, from small distal fruits to large proximal targets. Additionally, the instance density follows a long-tail distribution (max 73 objects per image), where high-density scenes inherently introduce severe mutual occlusion, providing a challenging benchmark for detection.

Beyond environmental statistics, the t-SNE visualization (Fig. 4) reveals the intrinsic difficulty of the task. Unripe pineapples form a loose,

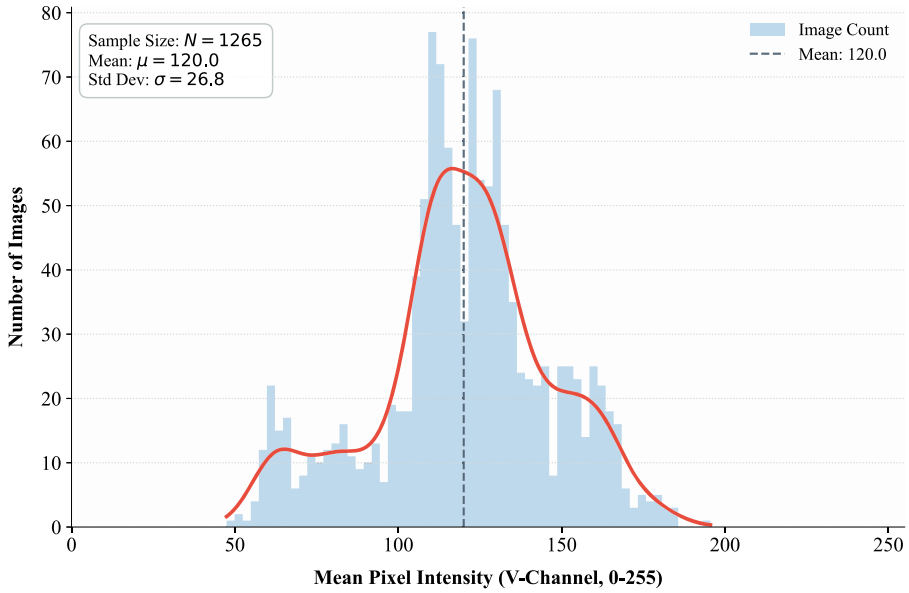

Fig. 3. Illumination Distribution Histogram. The V-channel intensity histogram ($\mu = 120.0, \sigma = 26.8$) spans an effective range of $[50, 185]$, demonstrating diverse natural lighting conditions preserved by rigorous manual screening.

Fig. 4. Visualization of features extracted by the detection model: (a) all samples, (b) unripe pineapple samples, and (c) ripe pineapple samples.

multimodal distribution, indicating high visual heterogeneity caused by the aforementioned variations in scale and angle. In contrast, ripe pineapples cluster tightly but share an overlapping boundary with unripe samples in the feature space. This overlap, driven by similar discriminative details, indicates that maturity assessment is susceptible to environmental interference. Therefore, the proposed SCRNet is specifically designed to handle these challenges by enhancing sensitivity to fine-grained features through spatial-channel reconstruction.

3.2. SCRNet for pineapple detection

This study proposes a multi-task object detection framework based on YOLOv11, named SCRNet, designed for pineapple detection and maturity assessment under complex field conditions. The overall architecture of SCRNet is illustrated in Fig. 5, consisting of three main components: the backbone, the neck, and the head. The backbone network is responsible for efficient feature extraction from the raw input images, effectively reducing redundant information while focusing on key regions relevant to fruit maturity. The neck structure is employed to fuse multi-scale feature representations, enhancing the model's capability to represent targets of varying scales and within complex backgrounds. The detection head leverages these features to output pineapple bounding boxes along with their maturity categories, achieving unified optimization of object localization and maturity assessment.

3.2.1. Design and implementation of the improved ScConv

Within the multi-task detection framework, although YOLOv11 offers excellent real-time performance, its standard convolution exhibits limited capacity for explicitly modeling spatial-channel dependencies. On the one hand, the fixed 3×3 convolutions in the early layers lack sufficient discriminative power for subtle local morphological textures and weak boundaries; on the other hand, the channel dimension lacks a

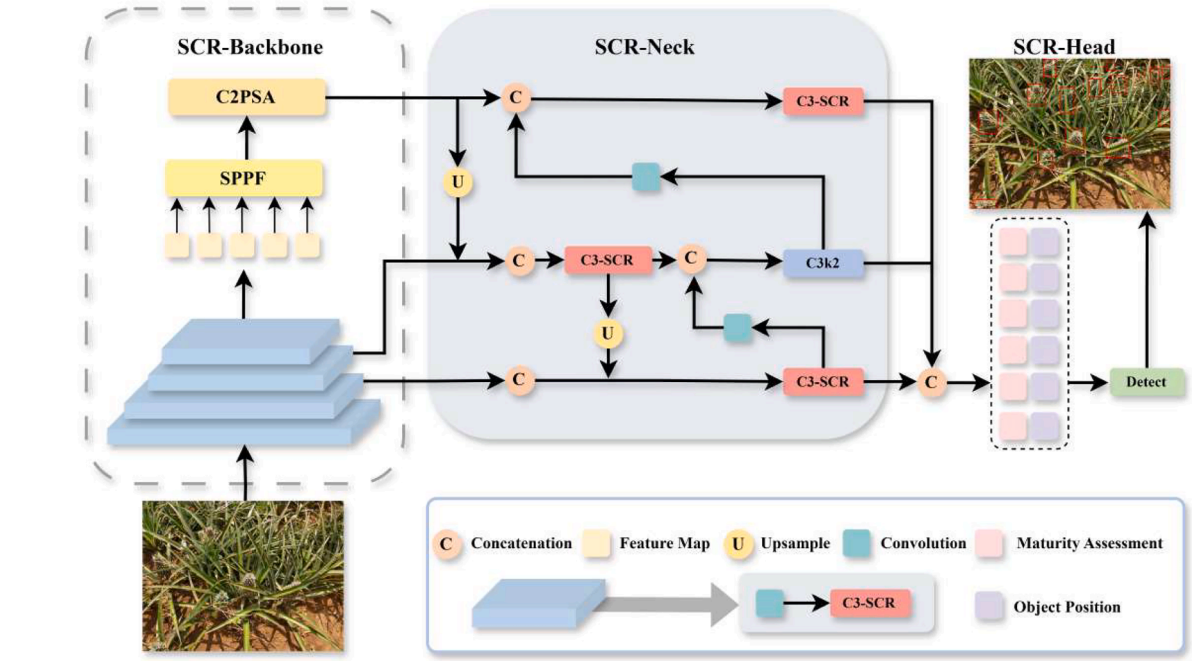

Fig. 5. Overall architecture of SCRNet.

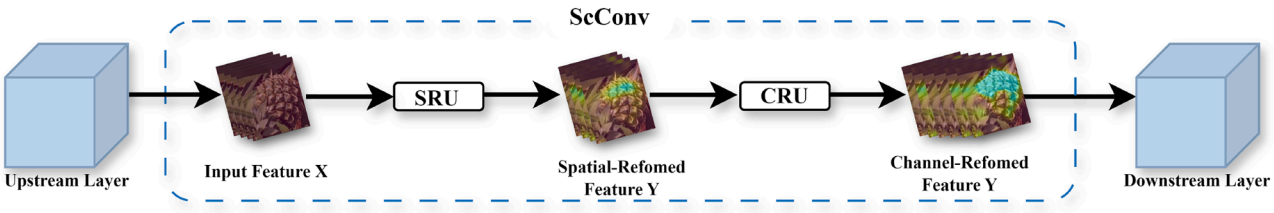

Fig. 6. Illustration of the ScConv module.

recalibration mechanism, causing feature representation to be uniformly distributed across all channels, which hinders focus on subspaces highly relevant to maturity assessment.

To address this, we introduce the improved ScConv (Li et al., 2023). Conceptually, distinct from traditional attention mechanisms (e.g., CBAM, SE) that operate on a "re-weighting" principle-multiplying features by learned masks while retaining the original redundant structure-ScConv operates on a "reconstruction" principle. It physically separates informative features from noise through a Split-Transform-Fuse paradigm, thereby achieving true redundancy reduction rather than mere suppression.

We retain its core mechanisms-the SRU's continuous gating based on Group Normalization and the CRU's "Divide-and-Conquer" channel fusion-while performing lightweight modifications to adapt it to the pineapple detection task. The workflow is illustrated in Fig. 6. Specifically, we implemented three targeted designs:

(i) Domain-Adaptive Hyperparameter Optimization: We defined the SRU fixed threshold, the CRU channel allocation coefficient α, and the group size g as shared adjustable hyperparameters. Unlike the original fixed implementation, we conducted systematic ablation experiments to maximize sensitivity to maturity-related features (e.g., skin color transitions) while maintaining robustness to high-variance field conditions such as changing illumination and occlusion.

(ii) Hardware-Friendly Alignment: We adjusted the CRU channel compression and grouping to rely on integer division (rounding up when necessary). This modification resolves the quantization misalignment often encountered in edge deployment, guaranteeing that the optimized network functions reliably on orchard machinery without structural conflicts.

(iii) Modular Integration Strategy: Instead of a simple layer replacement, we designed a seamless embedding strategy. We integrated the improved ScConv into the bottleneck of the C3k2 unit (forming the C3-SCR module detailed in Section 3.2.4). This design enhances the extraction of fine-grained maturity features while preserving tensor shape consistency, ensuring no computational bottlenecks are introduced during inference.

After integrating ScConv, the extraction of key features is enhanced, strengthening the model's sensitivity to detail-based maturity discrimination cues. By transitioning from "passive feature weighting" to "active feature reconstruction," the network can more precisely emphasize feature channels relevant to pineapple maturity, thereby improving its suitability for joint localization and maturity assessment.

3.2.2. *Spatial reconstruction unit (SRU)*

The workflow of the SRU is illustrated in Fig. 7. The objective of the SRU is to suppress spatial redundancy in intermediate features and highlight discriminative details without introducing significant additional computational overhead. Unlike standard spatial attention mechanisms that rely on heavy convolutional layers to learn attention masks, SRU exploits the inherent statistical properties of feature maps to estimate their spatial informativeness. We hypothesize that feature maps with higher variance contain discriminative texture information (e.g., pineapple eyes and edges), while those with lower variance correspond to redundant background (e.g., blurred soil or sky). Specifically, it first

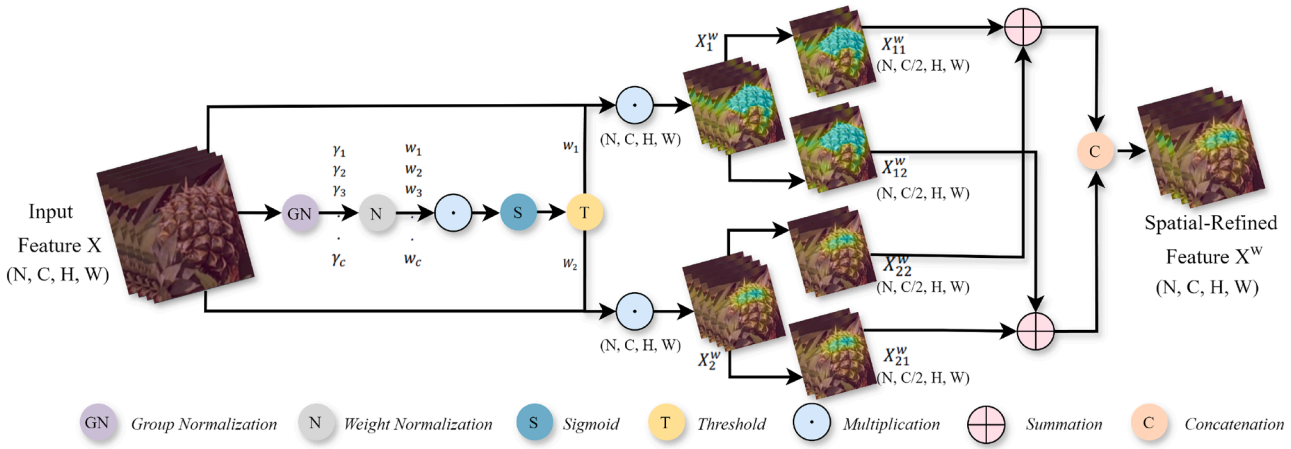

Fig. 7. Illustration of the spatial reconstruction unit (SRU).

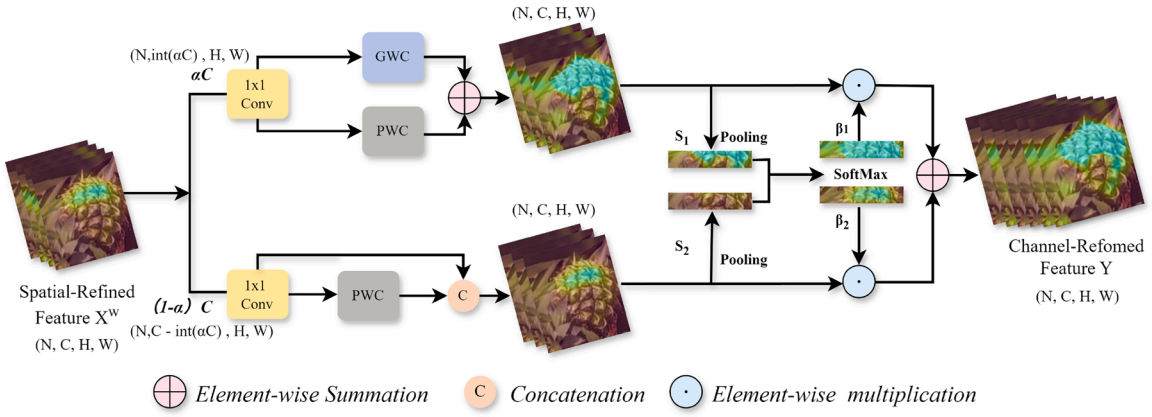

Fig. 8. Illustration of the channel reconstructing unit (CRU).

estimates the spatial information of each channel and applies gated binarization, followed by cross-channel interactions to enhance effective components while compressing redundant ones.

Given an input feature map $X \in \mathbb{R}^{N\times C\times H\times W}$, group normalization (GN) is applied. We utilize the learnable affine parameters of GN to measure the spatial importance of each channel:

$$Z = \mathrm{GN}(X) = \gamma \odot \frac{X-\mu}{\sqrt{\sigma^2+\varepsilon}} + \beta, \tag{1}$$

where in group normalization, we set the number of groups to 4, μ and σ denote the mean and variance within each group, respectively, ε is a small constant for numerical stability, γ and β are learnable parameters. The operator $\odot$ denotes element-wise multiplication (Hadamard product).

Crucially, the scaling factor γ acts as a gradient-aware indicator of channel deviation. Channels with significant spatial structures typically acquire larger γ values during training to preserve their activation variance. To estimate the spatial importance across channels, the scaling factors of the normalization layer are normalized by summation to generate a global spatial descriptor:

$$W_\gamma = \frac{\gamma}{\sum_{j=1}^{C}\gamma_j} \in \mathbb{R}^C. \tag{2}$$

Subsequently, Z and W_γ are applied element-wise and passed through a Sigmoid activation to obtain continuous responses. Instead of simple soft-weighting, we employ a "Split-and-Reconstruct" strategy to physically separate information from noise. A fixed threshold (set to 0.5) is then employed for gating: elements above the threshold are assigned hard weights, while those below retain soft weights, yielding two element-wise weight vectors W_1 and W_2. Accordingly, the features are separated into two branches:

$$X_i^w = X \odot W_i, \quad i \in \{1,2\}. \tag{3}$$

This operation effectively functions as a spatial filter. The "Informative" branch (X_1^w) retains features with high spatial variance (e.g., pineapple textures), allowing strong gradient flow, while the "Redundant" branch (X_2^w) suppresses background noise (e.g., leaves), thereby reducing the interference of complex field environments on maturity assessment.

For interaction, each branch is equally partitioned along the channel dimension and cross-summed to reinforce complementary information, resulting in two reconstructed features $\tilde{X}_1$ and $\tilde{X}_2$. Finally, the two are concatenated along the channel dimension to produce the spatially enhanced output:

$$X^w = \tilde{X}_1 \cup \tilde{X}_2. \tag{4}$$

3.2.3. Channel reconstructing unit (CRU)

To further suppress redundancy and enhance complementarity along the channel dimension, the Channel Reconstructing Unit (CRU) adopts a "Split–Transform–Fuse" strategy (Fig. 8). Standard channel attention mechanisms typically compress global spatial information into a single scalar using global average pooling (GAP). While efficient, this aggressive compression discards critical localized morphological cues (e.g., the texture of pineapple eyes) critical for maturity assessment. To address

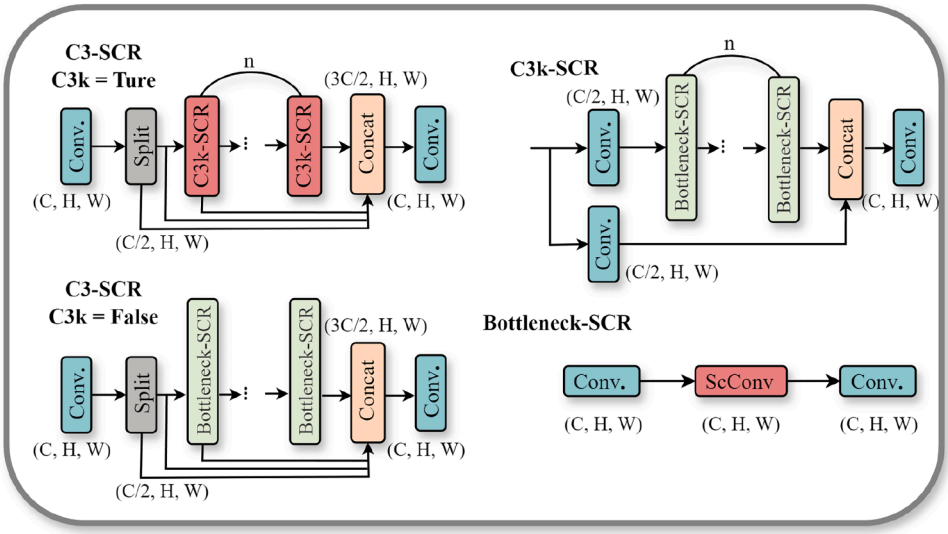

Fig. 9. Illustration of C3-SCR, C3k-SCR, and Bottleneck-SCR modules.

this, CRU introduces a "Divide-and-Conquer" approach that decouples high-level semantics from low-level details. First, the channels are proportionally divided into two branches and lightly squeezed. The upper branch applies grouped convolution and pointwise convolution in parallel to refine representative features, while the lower branch preserves detailed information and introduces residual connections to increase diversity. Finally, global pooling and softmax are used to perform channel-wise soft fusion.

Let the splitting ratio be $\alpha \in [0, 1]$ (set to 0.6), and the compression ratio be r (set to 2). Using 1×1 convolutions P_1 and P_2, the input X^w is mapped as:

$$X^{up} = P_1(X^w) \in \mathbb{R}^{\frac{\alpha C}{r} \times H \times W}, \tag{5}$$

$$X^{low} = P_2(X^w) \in \mathbb{R}^{\frac{(1-\alpha)C}{r} \times H \times W}. \tag{6}$$

To ensure the effectiveness of grouped convolution and avoid training–deployment inconsistency, the channel number after squeezing is subjected to group-divisible alignment.

The core innovation lies in the distinct transformation objectives of the two branches. In the upper branch, grouped convolution (GWC, group size g (set to 2), kernel size k (set to 3) and pointwise convolution (PWC) are applied in parallel to encode high-level semantic context with reduced parameters:

$$Y_1 = M_G X^{up} \oplus M_{P1} X^{up}, \tag{7}$$

while the lower branch extracts and preserves detailed information via 1×1 convolution and concatenates it with the original features to maintain shallow feature fidelity:

$$Y_2 = M_{P2} X^{low} \cup X^{low}. \tag{8}$$

Subsequently, global average pooling is applied to both branches to generate channel descriptors S_1 and S_2. Unlike independent sigmoid activation used in traditional attention, we employ a channel-wise soft competition mechanism to adaptively aggregate features. The fusion weights β_1 and β_2 are derived as:

$$\beta_k^c = \frac{e^{S_k^c}}{e^{S_1^c} + e^{S_2^c}}, \quad k \in \{1, 2\}, \tag{9}$$

where S_k^c denotes the global descriptor for the cth channel of branch k. This satisfies $\beta_1 + \beta_2 = \mathbf{1}$ for each channel.

The final CRU output is computed as:

$$Y = \beta_1 \odot Y_1 \oplus \beta_2 \odot Y_2. \tag{10}$$

Through this design, the model dynamically recalibrates the importance of semantic context (Upper) versus local detail (Lower) based on the input image content, offering a significant robustness advantage over static convolution operators. The input and output of CRU preserve identical dimensions (C, H, W), ensuring natural compatibility with residual and concatenation connections in the backbone.

3.2.4. C3-SCR module

To enhance the feature representation capacity of the YOLOv11 backbone in complex agricultural scenarios, this study introduces a structural optimization of the C3k2 module. Although the conventional C3k2 can effectively aggregate multi-scale features, its ability to capture critical local morphological information is limited by constraints in spatial and channel representations. Specifically, standard bottlenecks rely on content-agnostic convolution operations, which process the pineapple's textured surface and the surrounding sharp leaves indiscriminately. This lack of spatial adaptivity leads to the accumulation of background noise in deep layers. To address this issue, we propose the C3-SCR module, as illustrated in Fig. 8, by integrating the ScConv mechanism into the CSP bottleneck structure.

The C3-SCR module redefines the gradient flow to balance efficiency and feature refinement. Let the input feature map be X_{in}. Following the CSP topology (top-right of Fig. 9) and ensure the residual connection is consistently enabled in all units. X_{in} is partitioned into two subspaces via parallel 1×1 convolutions:

$$X_{main} = \mathrm{Conv}_1(X_{in}), \quad X_{short} = \mathrm{Conv}_2(X_{in}). \tag{11}$$

The processing branch X_{main} passes through a stack of n (set to 2) Bottleneck-SCR blocks. Unlike the standard bottleneck, we replace the second convolution with the Improved ScConv (bottom-right of Fig. 8). For the ith bottleneck ($i = 1, \ldots, n$):

$$Y_i = \mathrm{ScConv}\big(\sigma(\mathrm{Conv}_{\mathbf{3\times3}}(Y_{i-1}))\big) + Y_{i-1}, \tag{12}$$

where $Y_0 = X_{main}$, σ denotes the activation function ($\sigma = SiLU$), and $\mathrm{ScConv}(\cdot)$ encapsulates the sequential SRU and CRU operations defined in Sections 3.2.2 and 3.2.3.

By filtering spatial redundancy (SRU) and recalibrating channels (CRU) before the residual addition, Eq. (12) ensures that only refined, discriminative features are added to the identity flow, effectively preventing background noise accumulation across layers.

Finally, the outputs are fused with the shortcut branch via a transition layer:

$$X_{out} = \mathrm{Conv}_{1\times1}\big(\mathrm{Concat}(Y_n, X_{short})\big). \tag{13}$$

This integration enhances the module's capacity for fine-grained feature extraction and redundancy suppression, making it significantly more robust to the complex environments of natural farmland. Specifically, the C3-SCR module handles the computational trade-off by achieving superior feature representation power at a lower cost: the SRU reduces computational overhead by suppressing spatial redundancy using the learnable scaling factor (γ) to estimate channel importance for gating, thus enabling feature separation without relying on heavy convolution, while the CRU employs Grouped Convolution (GWC) and Pointwise Convolution (PWC) in parallel within its upper semantic branch to encode high-level semantic context. This structure leverages the parameter efficiency of GWC to achieve a significant reduction in parameter overhead and enhance the module's representational capacity. This structural optimization results in high inference speed, in line with the efficiency standards for consumer edge devices (Usman et al., 2025b).

It is worth noting that while this study implements the C3-SCR module within the YOLOv11 framework, its architectural design is generic and model-agnostic. The core mechanism of C3-SCR targets the fundamental redundancy in convolutional feature maps rather than specific artifacts of the YOLOv11 backbone. Structurally, the module is designed as a drop-in replacement for standard residual bottlenecks found in various one-stage detectors. For instance, it can seamlessly replace the C3 unit in YOLOv5 or the C2f unit in YOLOv8 by aligning the channel width and depth parameters. Therefore, the effectiveness of the spatial-channel reconstruction strategy is not strictly tied to YOLOv11 but is expected to generalize to other CSPNet-based or residual-based detection architectures.

3.2.5. Decoupled head for joint localization and maturity assessment

To effectively perform simultaneous localization and maturity assessment, SCRNet adopts a dual-stream optimization strategy. Architecturally, the decoupled head is replicated across three distinct feature pyramid network (FPN) levels, specialized for capturing multi-scale targets (small, medium, and large) through scale-specific receptive fields. Operationally, the model is trained using a robust data augmentation scheme comprising Mosaic and Mixup techniques. These strategies synthetically reconstruct training samples with diverse object densities and complex background noise, thereby preventing the model from overfitting to sparse scenarios and significantly enhancing its scale invariance and spatial robustness under dense clustering conditions. As shown in Fig. 10, unlike traditional coupled heads that share feature maps for all tasks, this design bifurcates the feature stream into two independent branches to resolve the potential feature conflict between regression and classification tasks.

The localization branch focuses on geometric features, such as edges and contours, to regress the bounding box coordinates. It utilizes CIoU Loss to ensure precise boundary alignment even in occluded scenarios. The mathematical form of the CIoU Loss is defined as:

$$L_{CIoU} = 1 - \mathrm{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v, \tag{14}$$

where IoU is the Intersection over Union, the second term measures the distance between the predicted box center b and the ground truth center b^{gt}, c is the diagonal distance of the smallest enclosing box covering both, and the term αv penalizes aspect ratio inconsistencies.

The maturity classification branch focuses on discriminative features, such as skin texture and color transition, to predict the probability distribution over maturity classes. This branch is optimized using Binary Cross-Entropy Loss. The complete BCE loss function, averaged over a batch of N samples, is given by:

$$L_{BCE} = \frac{1}{N}\sum_{i}^{N} -[w \cdot y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)], \tag{15}$$

where y_i is the ground truth label and p_i is the predicted probability for the ith sample in the batch. To address the severe class imbalance, a weighting factor w is introduced to the positive samples (ripe pineapples). We set w to be inversely proportional to the class frequencies (i.e., $w = N_{unripe}/N_{ripe}$), which significantly increases the loss contribution from the minority class and prevents the model from biasing toward the majority class.

This architectural decoupling is strategically vital for pineapple harvesting. Localization relies on structural information to determine where the pineapple is located, while maturity assessment relies on critical local morphological cues to determine the pineapple's status. By separating these optimization objectives, the model prevents feature conflict, ensuring that the subtle visual indicators of maturity are learned independently from spatial positioning.

4. Results and discussion

4.1. Evaluation metrics

This study employs *Precision (P)*, *Recall (R)*, and *F1 Score (F1)* as primary metrics for evaluating maturity assessment performance. Localization quality is quantified using AP_{50}, defined as the area under the *P-R* curve at IoU = 0.5. These metrics derive from fundamental components of the confusion matrix computed for maturity categories: *True Positives (TP)* represent the number of pineapples correctly identified as belonging to the maturity category, *False Positives (FP)* denote the number of ripe pineapples incorrectly classified as belonging to that maturity level, and *False Negatives (FN)* correspond to the number of pineapples whose maturity is undetected or misassessed.

From a pineapple harvesting perspective, the *P*-value reflects the degree of misjudgment in maturity assessment. An excessively low *P*-value may lead to premature harvesting of unripe pineapples or delayed harvesting of ripe ones. *R* represents the proportion of missed ripe fruits, directly impacting the harvesting efficiency of robotic harvesters. *F1* balances the metrics of *P* and *R*. A higher *F1 score* indicates more reliable maturity determination, demonstrating the model's robustness and generalization capability across varying field conditions. AP_{50} correlates with positioning accuracy: higher scores mean the model locates pineapples more precisely, ensuring robotic harvesters execute cutting operations accurately without damaging the fruit. These metrics provide a comprehensive evaluation of maturity assessment and positioning, critical for the practical deployment of intelligent pineapple harvesting systems. The specific evaluation metric formula is as follows:

Precision (P): Measures the proportion of correctly predicted positive samples among all predicted positive samples, defined as:

$$P = \frac{TP}{TP + FP}. \tag{16}$$

Recall (R): Measures the proportion of correctly predicted positive samples among all actual positive samples, defined as:

$$R = \frac{TP}{TP + FN}. \tag{17}$$

F1 Score (F1): The harmonic mean of precision (P) and recall (R), providing a balanced measure of a model's accuracy. It is calculated as:

$$F1 = 2 \times \frac{P \times R}{P + R}. \tag{18}$$

mean Average Precision (mAP): Evaluates the combined performance of precision and recall across different thresholds, defined as:

$$mAP = \frac{1}{C}\sum_{c=1}^{C}\int_{0}^{1} P_c(R), dR, \tag{19}$$

where $P(R)$ denotes the precision–recall curve. mAP is simply the arithmetic mean of the AP values across all classes. mAP_{50} and mAP_{75} indicate the average precision at an IoU threshold of 0.5 and 0.75. mAP_{50-95} denotes the mean Average Precision computed as the average of mAP values across 10 (IoU) thresholds, ranging from 0.50 to 0.95 in increments of 0.05

In addition, considering that the model needs to be deployed on actual edge computing devices, we use the number of *parameters* as a metric to measure model complexity, measured in millions, and evaluate inference speed in*frames per second (FPS)*.

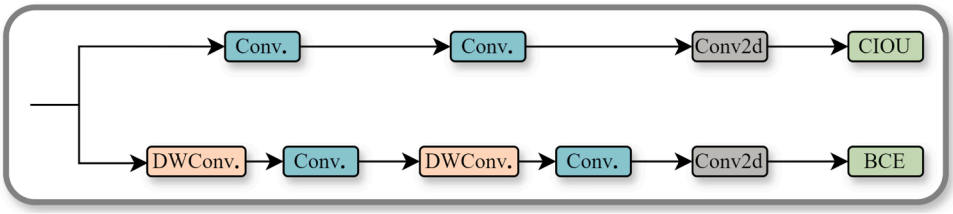

Fig. 10. Illustration of detect head.

Table 2
Performance metrics for different numbers of groups. The number of groups determines how many subsets the input channels are divided into for independent normalization in Group Normalization.

Number of groups	*P (%)* ↑	*R (%)* ↑	*F1 score (%)* ↑	mAP_{50} *(%)* ↑
1	85.060	81.800	83.398	88.844
2	86.347	82.235	84.241	88.898
4	85.594	**83.216**	**84.530**	88.983
8	86.082	81.404	83.689	88.662
16	**87.514**	79.330	83.239	**89.208**

4.2. Experimental setup

The experiments are conducted on the Ubuntu 22.04.2 LTS operating system. The hardware environment comprises an AMD Ryzen 9 5950X 16-Core Processor CPU and a GeForce RTX 3090 GPU with 24 GB of VRAM. The software environment is based on the PyTorch 2.6.0 deep learning framework, Python 3.9, and CUDA 12.4. All comparison algorithms are reproduced on the MMDetection (Chen et al., 2019) and Ultralytics (Jocher & Qiu, 2024) benchmark platforms. Hyperparameter settings and result validation criteria are standardized to ensure fairness in algorithmic comparisons. To ensure fair training, all comparison models are fine-tuned to use the same set of critical hyperparameters and anchors where applicable. The input image size is set to 640×640. During training, the stochastic gradient descent (SGD) optimizer is used with a momentum of 0.9 and a weight decay of 0.0005. All models are trained for 300 epochs with a batch size of 16 to guarantee full convergence. The checkpoint corresponding to the highest AP_{50} on the validation set is selected as the final model. The training utilizes a composite loss function comprising the CIoU Loss for bounding box localization, and the Binary Cross-Entropy (BCE) Loss for both objectness and classification tasks. The learning rate is controlled by a Cosine Annealing decay schedule. All reported results are presented as the mean and standard deviation of the best-performing round of AP50 metrics from three independent experiments, ensuring the robustness of statistical outcomes. All specific hyperparameter values and configurations adhere to the standardized settings of the Ultralytics platform for fair comparison; the confidence threshold is set to 0.6.

4.3. Ablation study and results discussion

4.3.1. Ablation study on the number of groups in SRU

In the SRU, a feature grouping mechanism was introduced during the normalization stage. Specifically, the number of groups determines how input channels are partitioned in Group Normalization, with each group undergoing independent normalization. When the group size is too small, normalization becomes overly coarse, limiting the ability to capture channel-wise subtle inter-channel differences. Conversely, an excessively large group size disperses channel statistics, weakening overall correlations. To investigate this effect, we conducted comparative experiments under identical training strategies and hyperparameter settings, varying the group size to 1, 2, 4, 8, and 16.

The experimental results are presented in Table 2. When the number of groups was set to 1, the method effectively reduced to Layer Normalization, achieving *Average Precision* of 88.844% and *F1 score* of 83.398%. Increasing the group size to 2 and 4 led to performance improvements, with the best results obtained at 4 groups: AP_{50} reached 88.998% and *F1 score* improved to 84.334%. This demonstrates that moderate grouping better captures inter-channel differences during normalization, thereby enhancing the representational capacity of spatial features.

However, performance declined as the group size increased further. At 16 groups, although *Precision* peaked at 87.514%, *Recall* dropped markedly to 79.33%, reducing the overall *F1 score* to 83.239%. This suggests that excessive grouping leaves too few channels per group, yielding insufficient normalization statistics. As a result, feature correlations weaken, which undermines detection stability and generalization.

In summary, selecting an appropriate group size in normalization plays a critical role in improving spatial feature modeling and detection performance. In this study, a group size of 4 achieved the best trade-off between *Precision* and *Recall*, validating the effectiveness of the proposed design.

4.3.2. Ablation study on the hyperparameter α in CRU

In C3k–SCR, the CRU module uses the hyperparameter α to control the channel allocation ratio between the "high-response" and "low-response" paths: αC channels preserve stronger target saliency, while $(1-\alpha)C$ channels focus on fine-grained supplementation and background suppression. These two paths jointly determine detection performance through subsequent spatial/channel interactions and fusion; therefore, the setting of α directly affects the balance of feature representation and the final results.

A systematic ablation study was conducted over $\alpha \in \{0.1, 0.2, \ldots, 0.9\}$, with results presented in Table 3. When α is too small (≤ 0.3), the high-response channels are insufficient, limiting salient feature representation. Conversely, when α is too large (≥ 0.8), the low-response channels are excessively compressed, reducing background suppression and markedly decreasing *Recall*. The overall comparison indicates that α of 0.6 achieves the highest *F1 score* while also yielding the highest recall across all settings; α of 0.7 attains the highest AP_{50}, though its *F1 score* is slightly lower than that of α of 0.6. Considering a balance between accuracy and stability, α was fixed at 0.6 in subsequent experiments.

4.3.3. Ablation study on C3-SCR

To rigorously validate the effectiveness of the proposed reconstruction mechanism, we conducted a systematic comparative analysis on the YOLOv11-s baseline, involving two dimensions: (1) benchmarking against representative attention modules (SE and CBAM Hu et al., 2018;

Table 3
Ablation results of the hyperparameter α in CRU. α is a hyperparameter that determines the allocation ratio between salient-feature channels and detail-enhancement channels in the CRU.

α	P (%) ↑	R (%) ↑	*F1 score* (%) ↑	mAP_{50} (%) ↑
0.1	86.940	80.113	83.434	89.117
0.2	85.083	82.113	83.556	89.044
0.3	87.746	80.823	84.215	88.962
0.4	**88.447**	79.922	83.960	89.009
0.5	84.664	81.961	83.294	89.085
0.6	85.594	**83.216**	**84.530**	88.983
0.7	86.175	80.818	83.440	**89.166**
0.8	83.910	82.347	83.125	89.005
0.9	85.660	80.426	82.991	88.620

Woo et al., 2018) and modern dynamic and deformable convolution designs (Dynamic and LDConv Han et al., 2024; Zhang et al., 2024c); and (2) isolating the internal SRU and CRU to quantify their individual contributions.

As presented in Table 4, the comparative results reveal distinct characteristics of each mechanism. The Dynamic Convolution strategy, while aiming to increase model capacity, results in a substantial rise in parameters, making it less ideal for storage-constrained edge devices. LDConv introduces flexible sampling shapes with a slight reduction in parameters, yet it yields a lower F1 score compared to the baseline on our dataset. Attention mechanisms like CBAM provide marginal gains but increase the model size. In stark contrast, C3-SCR achieves the most favorable trade-off: it delivers the highest detection performance while reducing the parameter count to the lowest level. This empirically confirms that our spatial-channel reconstruction strategy effectively reduces redundancy without compromising accuracy, outperforming both reweighting and dynamic aggregation approaches.

Furthermore, the internal ablation confirms the synergistic effect of our design. Integrating SRU or CRU individually yields improvements, but their combination in C3-SCR achieves the optimal balance, delivering superior overall detection performance.

4.3.4. Ablation study of the loss function

To evaluate the impact of bounding box regression loss on detection performance, we compared four loss functions: IoU, DIoU, GIoU, and CIoU, under the same training and evaluation protocols. The results are presented in Table 5. It can be observed that CIoU achieves the best overall performance: its *F1 score* reaches 84.342% (the highest among all settings), while also attaining the highest *Precision* and AP_{50} close to optimal. GIoU achieves the highest *Recall* but relatively lower Precision, resulting in *F1 score* inferior to that of CIoU. Although IoU achieves the highest AP_{50}, its balance between *Precision* and *Recall* is worse than that of CIoU. Overall, by simultaneously considering overlap, center distance, and aspect ratio, CIoU better aligns with bounding box alignment and shape constraints in detection tasks, providing a more robust trade-off on this dataset. CIoU was therefore adopted as the default bounding box regression loss in subsequent experiments.

4.4. Comprehensive performance analysis

In this section, we will analyze the model's performance from multiple perspectives: comparing its complexity and performance against the baseline model, evaluating its performance against current mainstream detection models, and analyzing the confusion matrix of SCRNet.

4.4.1. Performance-complexity benchmark comparison

To systematically evaluate the accuracy-efficiency trade-off of the proposed method, this section compares SCRNet with different-sized YOLOv11 models (n/s/m/l/x) on our self-built Pineapple dataset under unified data partitioning, training strategies, and post-processing configurations. As shown in Table 6, Figs. 11 and 12, we evaluate five different variants of YOLOv11 models and SCRNet on the Pineapple Dataset, incorporating the proposed C3-SCR module into YOLOv11-s. We build all five YOLOv11 variants entirely from the official YOLOv11 code. Model scale (depth and width) increases sequentially from n to x, theoretically yielding higher accuracy but slower inference speeds. Considering the model's deployment on edge computing devices for robotic arms, we avoid selecting models with more parameters and higher accuracy. Comparing all YOLOv11 variants clearly demonstrates the SCR module's enhancement effect: while reducing *parameters* by 0.5 million compared to YOLOv11-s, SCRNet achieves the best performance on the most critical metrics (AP_{50} and *F1 score*). Another interesting observation is that while larger parameter counts do yield higher accuracy in object detection tasks, the improvement in metrics is incremental and comes at the cost of substantial computational overhead. This makes such models unsuitable for practical deployment on edge devices like robotic arms.

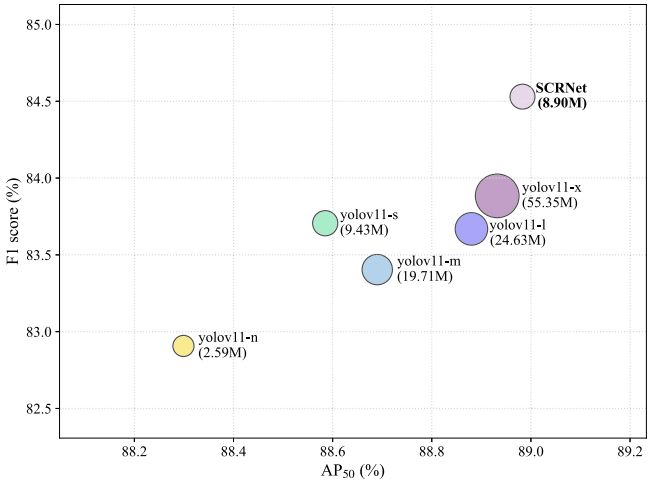

Fig. 11. Performance-Complexity Benchmark Comparison. The chart visualizes the trade-off between model performance and complexity. The x-axis and y-axis represent the AP_{50} *(%)* and *F1 score (%)*, respectively. The size of each bubble is proportional to the number of model *parameters* (in millions).

4.4.2. Analyzing the experimental results of the comparison to dominant methods

To objectively evaluate the effectiveness of the proposed method, we conducted a horizontal comparison with multiple representative detectors on the self-constructed pineapple dataset. All models were reproduced under consistent data splits, training procedures, and post-processing settings. The confidence threshold is set to 0.6.

As shown in Table 7 and Fig. 13, SCRNet achieves leading overall accuracy. Specifically, SCRNet demonstrates superior performance in localization accuracy, achieving the highest scores among all compared methods across the AP_{50}, AP_{75}, and $AP_{50:95}$ metrics. Although Precision (85.594%) and Recall (83.216%) are not the absolute highest, they are the most balanced among all models, with a minimal difference of only 2.378 percentage points. This excellent balance between P and R is the primary reason SCRNet achieves the leading *F1 score*, thus strongly demonstrating its robust generalization capability for maturity assessment.

Among the single-stage detection frameworks, YOLOv12 achieves an *F1 score* of 83.640% and an AP_{50} of 88.836%, placing it in the second tier. YOLOv8 and YOLOv11 register *F1 scores* of 82.457% and 82.907%, respectively, showing stable performance but slightly inferior to SCRNet. The two-stage Faster R-CNN exhibits higher *Recall* but lower *Precision*, resulting in an *F1 score* of only 77.492%. The end-to-end DDQ method does not show advantages under the scale and scenario of this dataset, while having a relatively large parameter count.

Table 4
Ablation study of C3-SCR and comparison with attention mechanisms on the Pineapple Dataset. Comparison includes internal component isolation and external state-of-the-art attention methods.

Method	*P (%)* ↑	*R (%)* ↑	*F1* (%) ↑	mAP_{50} (%) ↑	mAP_{75} (%) ↑	mAP_{50-95} (%) ↑	*Params* (M) ↓
YOLOv11-s	87.009	80.641	83.706	88.585	53.289	49.721	9.429
+ SE (Hu et al., 2018)	85.493	81.529	83.464	**89.324**	54.719	49.843	9.472
+ CBAM (Woo et al., 2018)	86.683	80.671	83.569	89.062	55.594	49.084	9.773
+ Dynamic (Han et al., 2024)	85.853	82.174	83.973	88.667	54.690	50.429	11.932
+ LDConv (Zhang et al., 2024c)	86.097	80.274	83.082	88.386	52.915	50.232	9.324
+ SRU Only	84.410	82.915	83.656	88.785	55.904	49.822	**8.765**
+ CRU Only	83.830	81.443	82.619	88.728	53.256	49.614	8.917
+ **C3-SCR**	85.594	**83.216**	**84.530**	88.983	**55.124**	**50.696**	8.903

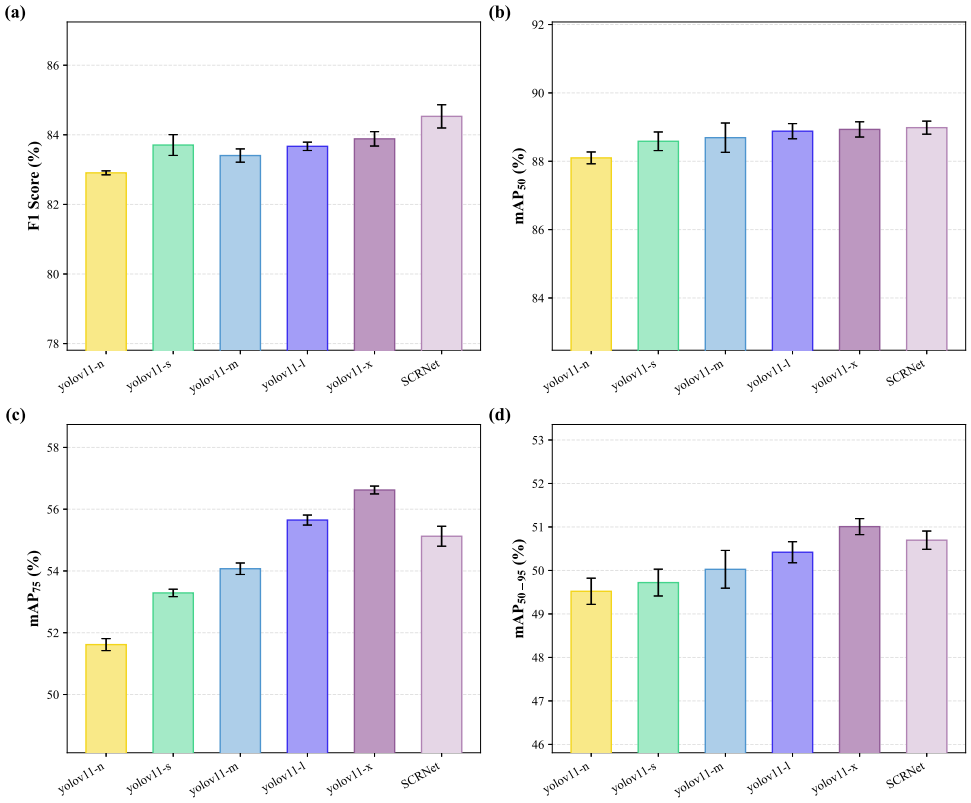

Fig. 12. Performance comparison of benchmark on the pineapple dataset.

Table 5
Ablation study of different bounding-box regression losses on the pineapple dataset.

Loss	*P (%)* ↑	*R (%)* ↑	mAP_{50} (%) ↑	*F1 score* (%) ↑
IoU	85.226	82.365	**89.278**	83.724
DIoU	85.223	80.793	88.421	82.987
GIoU	83.574	**83.676**	88.806	83.625
CIoU	**85.594**	83.216	88.983	**84.530**

In terms of efficiency, SCRNet achieves excellent performance with a moderate parameter count of 8.9 M, realizing a superior accuracy-complexity trade-off compared to many other detection models. This highlights that the C3-SCR module effectively achieves remarkable feature enhancement effectiveness at a low cost, simultaneously suppressing redundant information and enhancing feature extraction capabilities.

As shown in Table 8, we employs a paired samples *t*-test to conduct a significance analysis of performance metrics across different models, thereby assessing the statistical significance of differences between models. The paired samples *t*-test determines whether differences in model performance are statistically significant by examining whether the difference between the means of two related samples is significantly different from zero. For the *F1 score* and mAP_{75} metrics, all models exhibited statistically significant differences ($p < 0.05$) compared to the baseline model. Regarding the mAP_{50} metric, all models except YOLOv12 demonstrated significant differences relative to the baseline model. For themAP_{50-95} metric, the differences between YOLOv12, Faster RCNN, and RTMDet did not reach statistical significance, while the differences for all other models were significant. Overall, the proposed model exhibits statistically significant performance differences compared to the benchmark models across most key evaluation metrics, indicating that its performance improvement is statistically reliable and not attributable to random fluctuations.

As illustrated in Fig. 14, we select several representative algorithms for visualization. From the results, SCRNet demonstrates the most stable performance across both sides of the scene, with precise boundary alignment and consistent class prediction, exhibiting almost no missed detections in occluded or densely clustered regions and very few false positives. YOLOv12 ranks second, producing generally accurate bounding boxes, though occasional misses occur in dense immature areas, and some boxes are slightly undersized under backlight-

Table 6
Performance comparison of different model architectures on the pineapple dataset using updated metrics.

Model	*P (%)* ↑	*R (%)* ↑	*F1 score(%)* ↑	mAP_{50} *(%)* ↑	mAP_{75} *(%)* ↑	$mAP_{50:95}$ *(%)* ↑	*Params (M)* ↓
yolov11-n	86.110 ±0.366	79.934 ±0.210	82.907 ±0.057	88.099 ±0.173	51.616 ±0.194	49.521 ±0.301	**2.590**
yolov11-s	87.009 ±0.153	80.641 ±0.606	83.706 ±0.299	88.585 ±0.272	53.289 ±0.120	49.721 ±0.308	9.429
yolov11-m	87.092 ±0.594	80.031 ±0.843	83.404 ±0.191	88.690 ±0.429	54.072 ±0.186	50.026 ±0.434	19.714
yolov11-l	85.609 ±0.538	81.850 ±0.714	83.669 ±0.120	88.880 ±0.222	55.648 ±0.162	50.419 ±0.242	24.632
yolov11-x	**87.589** ±0.154	80.487 ±0.318	83.884 ±0.207	88.932 ±0.223	**56.621 ±0.127**	51.008 ±0.184	55.352
SCRNet	85.594 ±0.190	**83.216 ±0.358**	**84.530 ±0.334**	**88.983 ±0.191**	55.124 ±0.323	**50.696 ±0.210**	8.903

Table 7
Comparison results of different detection models on the pineapple dataset (Mean ± Standard Deviation over three runs).

Model	*P* (%) ↑	*R* (%) ↑	*F1 score* (%) ↑	mAP_{50} (%) ↑	mAP_{75} (%) ↑	mAP_{50-95} (%) ↑	*Params* (M) ↓
YOLOv8 (Jocher et al., 2023)	**88.252 ±0.202**	77.377 ±0.362	82.457 ±0.247	88.507 ±0.248	54.036 ±0.277	49.185 ±0.419	3.01
YOLOv10 (Wang et al., 2024b)	83.647 ±0.410	80.321 ±0.355	81.949 ±0.228	87.496 ±0.178	51.318 ±0.174	49.401 ±0.267	2.70
YOLOv11 (Jocher & Qiu, 2024)	86.110 ±0.366	79.934 ±0.210	82.907 ±0.057	88.299 ±0.173	54.072 ±0.186	49.521 ±0.301	2.59
YOLOv12 (Tian et al., 2025)	86.139 ±0.506	81.291 ±0.580	83.640 ±0.311	88.836 ±0.152	52.247 ±0.202	50.162 ±0.285	**2.52**
Faster R-CNN (Ren et al., 2015)	71.036 ±0.607	85.258 ±0.644	77.492 ±0.096	86.782 ±0.518	51.756 ±0.295	50.414 ±0.210	41.54
ATSS (Zhang et al., 2020)	82.613 ±0.299	81.033 ±0.190	81.818 ±0.049	82.563 ±0.153	43.400 ±0.239	44.220 ±0.122	19.10
DDQ (Zhang et al., 2023)	81.332 ±0.315	77.984 ±0.193	79.623 ±0.226	79.788 ±0.333	44.397 ±0.101	44.167 ±0.224	48.31
PAA (Tran et al., 2022)	45.414 ±0.413	83.610 ±0.241	58.931 ±0.289	83.895 ±0.274	46.536 ±0.239	45.532 ±0.193	32.30
RTMDet (Lyu et al., 2022)	69.780 ±0.288	**86.562 ±0.074**	77.223 ±0.171	88.333 ±0.142	53.748 ±0.224	50.353 ±0.158	52.26
VFNet (Zhang et al., 2021)	59.278 ±0.370	80.910 ±0.165	68.456 ±0.147	77.042 ±0.264	49.361 ±0.221	47.811 ±0.161	32.26
SCRNet	85.594 ±0.190	83.216 ±0.358	**84.530 ±0.334**	**88.983 ±0.191**	**55.124 ±0.323**	**50.696 ±0.210**	8.90

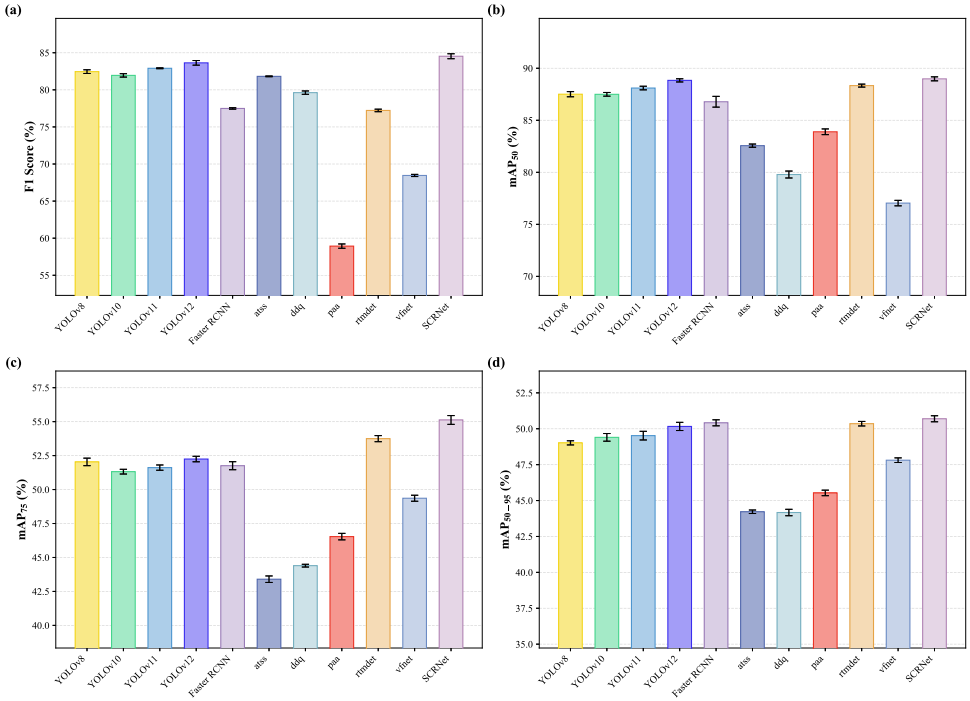

Fig. 13. Performance comparison of different detection models on the pineapple dataset.

Table 8
Significance test results (p-value) for different models. Bold indicates significant difference at $p < .05$.

Model	*F1 score*		mAP_{50}		mAP_{75}		mAP_{50-95}	
	t	*p*	*t*	*p*	*t*	*p*	*t*	*p*
YOLOv8 (Jocher et al., 2023)	13.63	**0.0053**	5.88	**0.0277**	9.18	**0.0117**	8.26	**0.0143**
YOLOv10 (Wang et al., 2024b)	8.81	**0.0126**	69.92	**0.0002**	20.72	**0.0023**	7.38	**0.0178**
YOLOv11 (Jocher & Qiu, 2024)	9.12	**0.0118**	4.40	**0.0479**	36.30	**0.0008**	7.15	**0.0190**
YOLOv12 (Tian et al., 2025)	4.36	**0.0488**	0.78	0.5189	9.60	**0.0107**	3.72	0.0654
Faster RCNN (Ren et al., 2015)	30.56	**0.0011**	6.47	**0.0231**	10.11	**0.0096**	1.17	0.3630
ATSS (Zhang et al., 2020)	14.13	**0.0050**	56.08	**0.0003**	43.69	**0.0005**	58.00	**0.0003**
DDQ (Zhang et al., 2023)	15.07	**0.0044**	55.93	**0.0003**	61.42	**0.0003**	50.36	**0.0004**
PAA (Tran et al., 2022)	78.55	**0.0002**	18.91	**0.0028**	27.62	**0.0013**	37.26	**0.0007**
RTMDet (Lyu et al., 2022)	36.65	**0.0007**	9.50	**0.0109**	25.78	**0.0015**	3.02	0.0942
VFNet (Zhang et al., 2021)	66.49	**0.0002**	46.65	**0.0005**	91.80	**0.0001**	102.44	**0.0001**

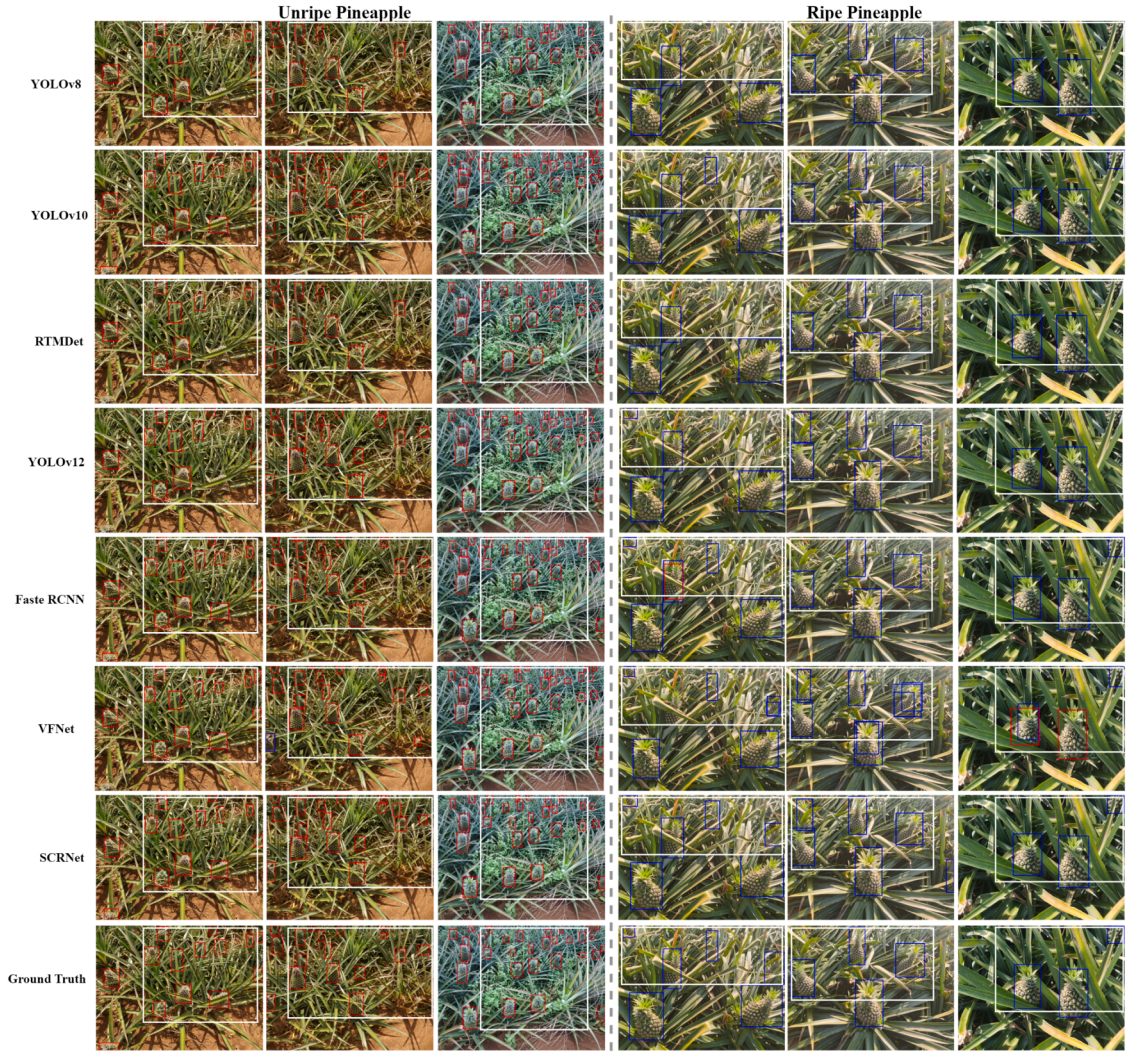

Fig. 14. Visualized object detection results of different models on the pineapple dataset. Qualitative comparison of different detectors in complex field scenarios. The dashed line marks the scene boundary: the left side contains ripe pineapples and the right side contains unripe ones. Bounding-box colors denote classes: red for unripe objects and blue for ripe objects.

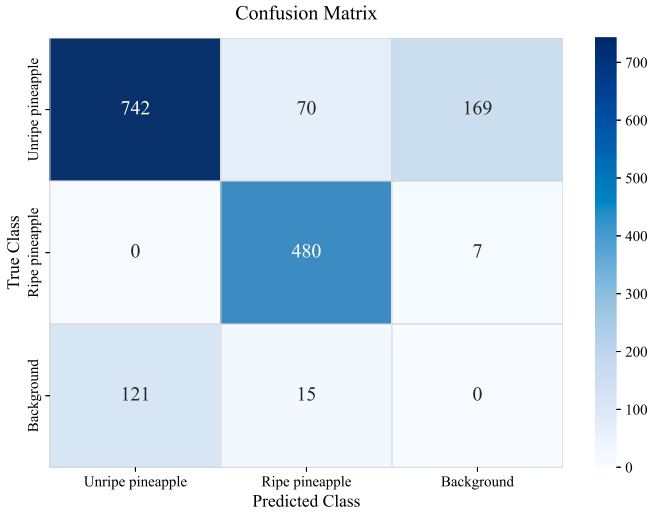

Fig. 15. Confusion matrix of pineapple maturity assessment.

ing. YOLOv8 and YOLOv11 show moderate performance, tending to miss targets under leaf occlusion or wide viewing angles; in regions with highly textured leaf veins, one or two false positives occasionally appear, and boundaries may be slightly loose. ATSS covers multiple scales but still misses targets in dense immature regions and occasionally produces false positives on sharp leaf textures. RTMDet achieves high recall but exhibits significant false positives and duplicate boxes, often misidentifying specular leaf highlights as objects. VFNet exhibits moderate robustness to complex backgrounds, resulting in more false positives and occasional cross-classification errors in Maturity regions. PAA has lower *precision* and is heavily affected by leaf textures, with prominent false positives and duplicate boxes. Faster R-CNN generates dense candidate boxes, which benefits small and occluded objects but increases redundancy and repeated detections, limiting *precision*. DDQ is sensitive to occlusion and small objects under the current dataset, resulting in noticeable missed detections. Overall, SCRNet outperforms all compared methods in both localization completeness and class stability, significantly reducing false positives and cross-class misclassification.

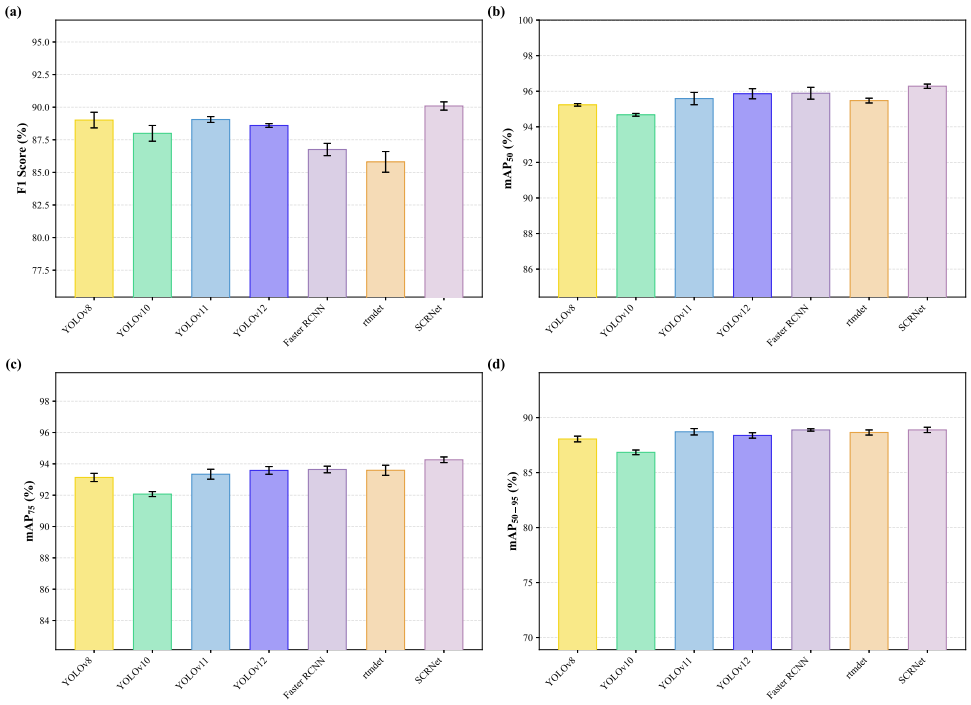

Fig. 16. Performance comparison of different detection models on the peach dataset.

4.4.3. Maturity assessment and error analysis

Given the subtle visual differences between pineapple maturity stages, which present a significant challenge for classification, this study provides a comprehensive analysis of SCRNet's misclassification patterns across these stages. shown in Fig. 15, a confusion matrix was employed to assess the model's maturity recognition performance and identify error sources.

From a maturity classification perspective, the model demonstrated high reliability. Out of 487 genuine mature samples, 480 were correctly identified, with no mature samples misclassified as immature. This confirms the model's ability to capture the key visual features of mature pineapples accurately. For the 981 genuine immature samples, 742 were correctly classified, while only 70 were erroneously predicted as mature. The low confusion between maturity stages indicates the model effectively handles the subtle distinctions between these two categories. However, error analysis reveals that the primary challenge lies in distinguishing targets from background. During detection, 136 background areas were mistakenly classified as pineapples, with 121 misclassified as unripe and 15 as ripe. This misclassification of background areas as pineapples primarily drives the increase in false alarm rates and precision decline. Such errors are often caused by background elements that resemble the color or texture of pineapples. The second major issue is target misses: 169 unripe pineapples and 7 ripe pineapples were incorrectly identified as background (false negatives). This misclassification significantly impacted the recognition accuracy of unripe pineapples. It is worth noting that the dataset was collected in real agricultural environments, where dense pineapple planting led to complex occlusions and overlaps. This made some unripe fruits, with weaker visual features, prone to being misclassified as background, leading to false negative errors. Qualitatively, we observed an operational limit under extreme occlusion. When the visible surface area of the fruit drops below approximately 30% (i.e., > 70% occlusion), the model struggles to extract sufficient textural features (e.g., pineapple eyes) for reliable detection. This indicates that severe obstruction remains a physical bottleneck for static single-view perception in dense fields.

In summary, the model effectively addresses the challenge of distinguishing between maturity stages. Future efforts should focus on reducing false positives in background detection and improving target detection in complex agricultural environments to minimize false negatives caused by occlusion.

4.4.4. Transfer evaluation on the peach dataset

To rigorously verify the generalization capability of SCRNet and alleviate concerns regarding potential domain overfitting to the specific pineapple dataset, we extended our evaluation to an external public benchmark: the Peach Maturity Dataset (Yuan, 2024). This dataset shares critical characteristics with pineapple harvesting, including the need to distinguish maturity stages based on subtle visual cues (color and texture) under complex occlusion and varying illumination. All models were retrained and tested under identical experimental protocols to ensure fair comparison. The quantitative results are summarized in Table 9.

As summarized in Fig. 16, SCRNet maintains its performance leadership across all key metrics. Specifically, it achieved a *Precision* of 93.206% and an *F1 score* of 90.091%, surpassing the latest YOLOv12 and the two-stage Faster R-CNN. Notably, SCRNet's mAP_{50} reached 96.283%, demonstrating its superior localization precision in unstructured orchard environments. The consistent performance gain on a different fruit variety underscores that the "active feature reconstruction" principle of our C3-SCR module effectively captures universal morphological cues (e.g., skin texture, color transitions) while suppressing environment-specific noise. This confirms that SCRNet possesses strong domain adaptability, making it a reliable tool for diverse selective harvesting applications.

4.5. Model performance test on edge devices

To comprehensively evaluate the practicality of models on resource-constrained devices, end-to-end performance testing is conducted on several YOLOv11 variants and the SCRNet model using the NVIDIA Jetson Nano development board. The Jetson Nano is a compact, low-power embedded AI computing module that is capable of efficiently executing modern AI algorithms on edge devices, making it well-suited for robotic arm applications. To provide a deeper insight into the model's applicability for selective harvesting, and drawing upon the robust validation

Table 9
Comparison results of different detection models on the peach dataset (Mean ± Standard Deviation over three runs).

Model	P (%) ↑	R (%) ↑	*F1 score* (%) ↑	mAP_{50} (%) ↑	mAP_{75} (%) ↑	mAP_{50-95} (%) ↑	*Params* (M) ↓
YOLOv8 (Jocher et al., 2023)	92.112 ±2.101	86.175 ±2.288	89.011 ±0.599	95.234 ±0.072	93.132 ±0.262	88.055 ±0.260	3.01
YOLOv10 (Wang et al., 2024b)	91.202 ±2.404	85.046 ±1.142	87.994 ±0.604	94.675 ±0.079	92.070 ±0.159	86.842 ±0.213	2.70
YOLOv11 (Jocher & Qiu, 2024)	90.687 ±0.502	87.481 ±0.620	89.054 ±0.223	95.588 ±0.346	93.338 ±0.317	88.713 ±0.289	2.59
YOLOv12 (Tian et al., 2025)	91.560 ±0.429	85.821 ±0.517	88.596 ±0.141	95.859 ±0.284	93.581 ±0.246	88.379 ±0.248	**2.52**
Faster R-CNN (Ren et al., 2015)	85.233 ±0.324	88.326 ±0.630	86.754 ±0.471	95.887 ±0.333	93.641 ±0.210	88.879 ±0.108	41.54
RTMDet (Lyu et al., 2022)	83.102 ±0.863	**88.690 ±0.713**	85.805 ±0.794	95.474 ±0.134	93.589 ±0.322	88.648 ±0.233	52.26
SCRNet	**93.206 ±0.480**	87.184 ±0.970	**90.091 ±0.317**	**96.283 ±0.119**	**94.260 ±0.181**	**88.882 ±0.242**	8.90

Table 10
Performance comparison of different models on the Jetson Nano edge device. Bold indicates the best performance in each column.

Model	P (%) ↑			R (%) ↑			*F1 score* (%) ↑	mAP_{50} (%) ↑	mAP_{50-95} (%)	*FPS* ↑
	unripe	ripe	all	unripe	ripe	all				
yolov11-n	83.245	85.071	84.158	74.336	83.051	78.694	81.388	85.356	48.443	**76.9**
yolov11-s	84.060	**90.750**	87.405	76.641	83.139	79.890	83.548	86.322	49.966	65.4
yolov11-m	**87.826**	87.477	**87.652**	72.140	82.875	77.508	82.275	87.201	50.598	42.7
yolov11-l	82.980	84.282	83.631	**77.939**	**87.857**	**82.898**	83.257	87.986	51.001	38.0
yolov11-x	84.861	84.940	85.092	75.344	84.861	80.103	82.527	**88.512**	**51.296**	27.4
SCRNet	84.611	87.385	85.998	76.384	86.441	81.413	**83.641**	87.332	50.813	53.2

Fig. 17. Pineapple detection and maturity assessment visualization in images captured by edge devices. The left side contains ripe pineapples, and the right side contains unripe ones. Bounding-box colors denote classes: red for unripe objects and blue for ripe objects.

framework for machine instinctive vision (Khan et al., 2024b), we expanded the evaluation metrics to include detailed *Precision* and *Recall* across different maturity stages, alongside the overall *F1 score*, *Average Precision* (AP_{50}, AP_{50-95}), and inference speed measured in *Frames Per Second (FPS)*.

Table 10 compares the performance on edge hardware. While detection accuracy generally scales with model size, the heavy YOLOv11-x is limited by low inference speed. Conversely, the lightweight YOLOv11-n compromises detection reliability. Although YOLOv11-l achieves the highest recall, its lower inference speed restricts deployment efficiency. SCRNet overcomes these trade-offs, achieving the best overall *F1 score* and a ripe recall that significantly surpasses both the baseline YOLOv11-s and the larger YOLOv11-x. Consequently, SCRNet achieves an exceptional balance across localization accuracy, maturity assessment capability, and inference FPS, demonstrating superior overall performance required for efficient robotic harvesting. From an end-to-end deployment perspective, the achieved inference speed of 53.2 FPS (approximately 19ms latency) comfortably exceeds the typical frequency requirements for real-time visual servoing (20–30 Hz). Given that the physical execution of the harvesting cycle, comprising arm manipulation, gripping, and separation, is mechanically limited and typically spans several seconds per fruit, the proposed vision system is sufficiently fast to ensure it does not become a bottleneck in the overall operation.

Furthermore, to achieve the closed-loop process of "localization-maturity assessment-selective harvesting," we conduct systematic field validation by processing real pineapple images directly on edge devices in actual agricultural environments. When a pineapple is detected and localized, and its maturity assessment indicates maturity, it signifies that the pineapple is ready for harvest. To ensure the authenticity and reliability of experimental results, we specifically retain certain pineapple samples from the test field for on-site testing. As shown in Fig. 17, we collect multiple pineapple samples at both immature and mature stages within a defined area and conduct field tests. We deploy the model on edge computing devices to perform simultaneous pineapple localization and maturity assessment, subsequently relaying this critical decision data to the harvesting machinery. This workflow simulates the coordination between visual detection and the robotic manipulator's grasping process. As shown in Table 11, to substantiate the system's practical applicability in real-world harvesting, we conduct a statistical validation on a sample set of 530 pineapples. Under our protocol, a harvesting action is triggered only when a target is successfully localized and classified as 'ripe'. Consequently, we define a successful harvest as the accurate conjunction of precise localization and correct maturity assessment. Our error analysis indicates that these failures are primarily attributed to severe occlusion.

In summary, the proposed method is validated across multiple dimensions. Comparative analyses with various YOLOv11 variants confirm its optimal balance between speed and accuracy. Moreover, real-device and real-world experiments verify its practical applicability and superiority in identifying pineapple locations and assessing maturity under real agricultural conditions.

Table 11
Empirical validation of robotic harvesting system performance on NVIDIA Jetson Nano. Failures are categorized into False Positives (misidentifying background as fruit), False Negatives (missing targets), and FDSR stands for Field Detection Success Rate.

Maturity Stage	Total Samples	Successful Harvest	False Positives	False Negatives	FDSR
Unripe	280	262	4	14	0.936
Ripe	250	238	2	10	0.952
Overall	530	500	6	24	0.943

5. Conclusions

This study addresses two major challenges in automated pineapple harvesting: maturity assessment and pineapple detection. We propose SCRNet, an efficient multi-task model based on one-stage detection, and construct a novel pineapple maturity dataset. Through the spatial-channel reconstruction module, SCRNet effectively enhances the feature representation of fruits at different maturity levels, achieving superior performance across multiple metrics. The released dataset covers diverse acquisition scenarios, providing a foundation for future research. Experimental results demonstrate that SCRNet significantly improves detection and maturity assessment accuracy while maintaining efficient inference, showing promising potential for practical automated harvesting applications. Nevertheless, the model still exhibits performance degradation under extreme occlusion or highly complex backgrounds. Future work will focus on further optimizing the model architecture and enhancing coordination with harvesting machinery to achieve more efficient and intelligent in-field fruit picking.

CRediT authorship contribution statement

Songwei Lian: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization; **Zhe Shan:** Writing – review & editing, Validation, Methodology, Supervision, Software, Conceptualization; **Ziming Yang:** Data curation, Visualization, Validation, Investigation, Formal analysis; **Lei Zhou:** Data curation, Methodology, Supervision, Conceptualization; **Cong Lin:** Supervision, Project administration, Data curation, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported in part by the Natural Science Foundation of Guangdong Province (2025A1515011356), the program for scientific research start-upfunds of Guangdong Ocean University (060302112405), the Undergraduate Innovation Team Project of Guangdong Ocean University (CXTD2024011 and JDTD2024003), Chinese Academy of Tropical Agricultural Sciences for Science and Technology Innovation Team of National Tropical Agricultural Science Center (NO.CATASCXTD202513).

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